Introduction to AI for Cybersecurity

Today's goal • 3/3 items

Module 1
Course Introduction

Module 2
AI for Cybersecurity Professionals

Module 3
Ham or Spam? Detecting Email Cybersecurity Threats with AI

Module 4
Securing User Authentication

Understanding and Securing User Authentication Threats

- Securing User Authentication
  Video • 5 min
- Reading References
  Reading • 15 min
- Understanding and Securing User Authentication Threats
  Practice Assignment • 15 min

Implementing Biometric Solutions: Keystroke and Facial Recognition

- Keystroke and Facial Recognition Biometric Authentication
  Video • 6 min
- Reading References
  Reading • 15 min
- Implementing Biometric Solutions: Keystroke and Facial Recognition
  Practice Assignment • 15 min

Module-end Assessments

- **Self-Reflective Reading: Authentication is Paramount**
  Reading • 40 min
- Securing User Authentication
  Graded Assignment • 1h
- Practice Lab: Detecting IoT Malware Behavior in Network Traffic
  Lab • 1h

Self-Reflective Reading: Authentication is Paramount

0:16 / 2:26 2x

Introduction

In today's digital landscape, user authentication security is paramount, especially as cyber threats become increasingly sophisticated. One of the most promising approaches to enhance security is using biometrics, which leverages unique biological characteristics of individuals—such as fingerprints, facial recognition, and iris scans—to verify identities. This literature review explores various applications of biometric technology in securing cyber systems, highlighting examples beyond those discussed in the lectures.

Tasks and Steps to Complete

1. **Literature Review on Biometrics:**
   - Research various types of biometric authentication methods and their applications in cyber security.
   - Identify at least two examples of biometric systems not covered in the lecture, such as voice recognition or palm vein scanning.
2. **Hands-On ML Examples:**
   - Access the provided virtual machine (VM) containing the Jupyter Notebook for Chapter 6.
   - Summarize the development of each machine learning (ML) analytic example mentioned in the notebook, using the language from the ML analytic development process.
   - Reflect on how an accurate AI fraud prevention tool could simplify a cyber analyst's role.
   - Examine the training and testing datasets before classification. Determine if the features are raw or have undergone a transformation, and provide an explanation.
   - Investigate the Python documentation for each classifier used in the examples. Identify performance tuning parameters and optimize these parameters to enhance classification outcomes.

Self-Reflective Questions

- **For the Literature Review:**
  - How does understanding different biometric systems contribute to the overall security framework in cyber systems?
  - What challenges might organizations face in implementing biometric authentication methods?
- **For the Hands-On ML Examples:**
  - How does the use of ML analytics enhance the effectiveness of cyber security measures?
  - In what ways can improved classification results impact the real-time detection of cyber threats?
  - How do feature transformations affect the performance of ML classifiers, and what insights can be gained from analyzing these transformations?

By addressing these tasks and reflective questions, learners will deepen their understanding of biometrics and machine learning's role in securing user authentication within cyber systems.

Go to next item ✓ Completed

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